Lab07-Template

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VNU-HCM, HCMUS, FIT, SE

CS202 - Programming Systems

25A01 - 02

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A: YY: 01

H: YY: 03

> 💡 Students are permitted to use AI assistance for this series of homework.
> If you do not use AI: Please include Declaration A in your README.md file: "I did not use AI for this homework."
> If you use AI: Please include Declaration B in your README.md file: "I did use AI for this homework." The expected workflow is that you prompt the AI, it generates a plan, you review that plan, the AI executes the action, and you review, edit, and re-prompt as needed. Ultimately, you bear full responsibility for the final result.

Design class diagrams using Mermaid format and implement the required functionality in C++ for the following assignments. For your final submission, please ensure you include the C++ source code, the raw Mermaid syntax, and the rendered image files for each class diagram.

Class diagram: https://en.wikipedia.org/wiki/Class_diagram

Use stereotypes to annotate the meaning of classes and methods.

```
<<interface>>, <<abstract>>, <<concrete>>, <<constructor>>,
<<destructor>>, <<friend>>, <<static>>, <<virtual>>, <<pure
virtual>>, <<override>>, <<operator>>.
```

```
Ex: +<<pure virtual>> getSalary(): doubl
```

Assignment 1 - My Vector

Define the following methods for your own class, MyVector, a class template.

In the main() function, create a vector of integers and a vector of fractions to test your defined methods.

Apply exception handling mechanism by throw exceptions in specific methods and try...catch... in the main() function.

```
template<class T>
class MyVector{
private:
    T *arr;
    int size;
public:
    // empty array
    MyVector();

    // n zeros
    MyVector(int n);

    MyVector(T *a, int n);
    MyVector(const MyVector &v);

    ~MyVector();

    int getSize();
    T getItem(int index);
    void setItem(T value, int index);
```

```
    void add(T value);
    void addRange(T *a, int n);
    void clear();
    bool contains(T value);
    void toArray(T *arr, int &n);
    bool equals(const MyVector &v);
    int indexOf(T value);
    int lastIndexOf(T value);
    void insert(T value, int index);
    void remove(T value);
    void removeAt(int index);
    void reverse();
    string toString();

    void sortAsc();
    void sortDesc();
};
```

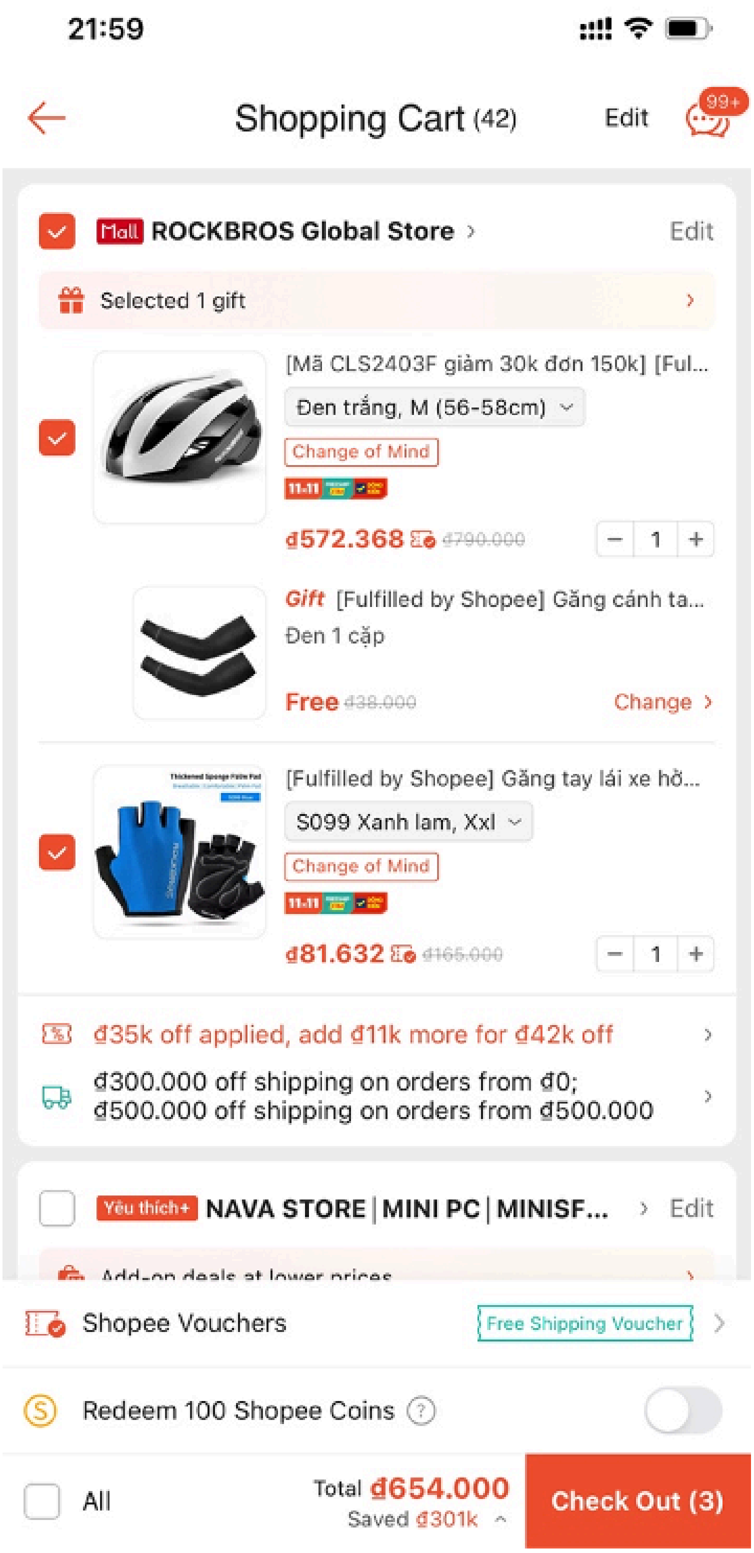
Assignment 2

Draw a class diagram and write a simple console application in C++ to simulate the following screens.

Write a report to explain your solution.

New requirements:

- Product information will be loaded from a text file.
- Orders will be saved to a text file.
- Support at least one type of promotion. For example: The system will offer a 10% discount that applied only if the total bill is equal or greater than 200,000 VND.



Assignment 3

Write a presentation (Markdown format) with at least 20 slides to explain the functional programming concepts in Python.